Statistics, Counting & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

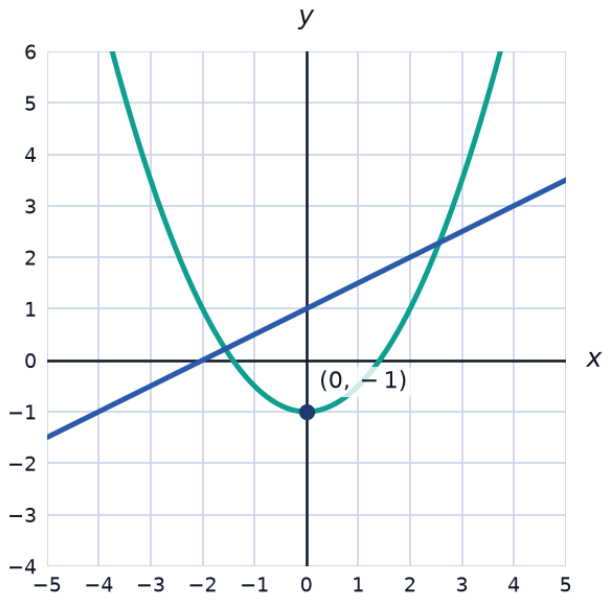
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 5 red and 3 blue balls. Find the probability that 3 balls drawn without replacement are all red.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $P(A) = 0.6$ and $P(B|A) = 0.4$, find the joint probability $P(A \cap B)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box contains 3 red, 4 white, and 5 blue cards. Find the probability of drawing a card that is not white.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple (LCM) of 12 and 18.

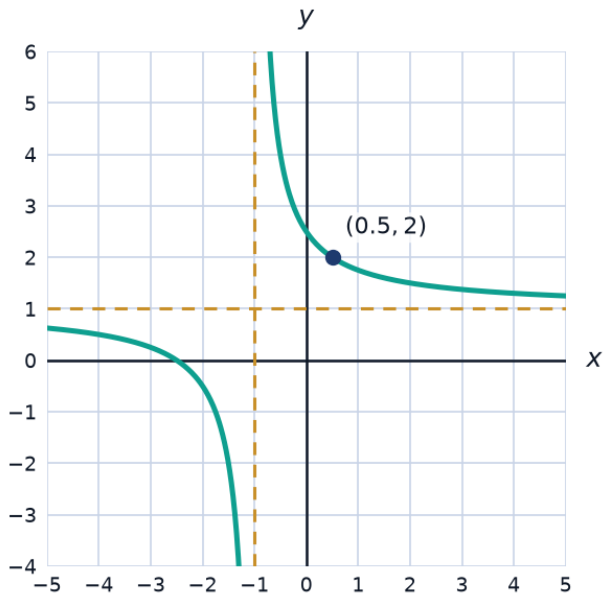
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical asymptote(s) of $f(x) = \frac{3x}{x^2 - 4}$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\lim_{x \to \infty} \frac{3x^2 + 1}{x^2 - 2}$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set has standard deviation 3 and mean 20. If every value in the data set is increased by 5, find the new mean and new standard deviation.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the finite series $\sum_{k=1}^{4} (2k - 1)$.

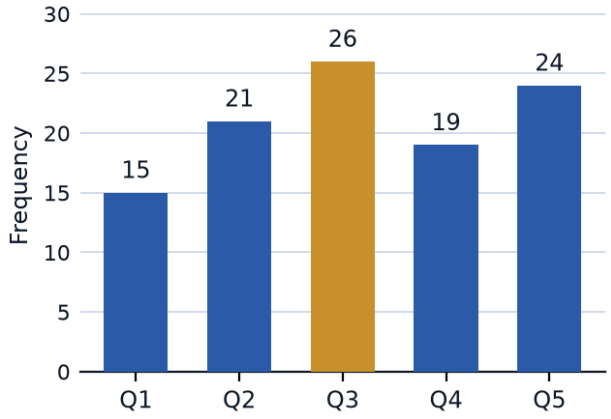
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Expand $(a + b)^5$ and find the coefficient of the term a^2b^3.

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10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The mean of 10 numbers is 15. One number is removed and the new mean of the remaining 9 numbers is 14. Find the removed number.



11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

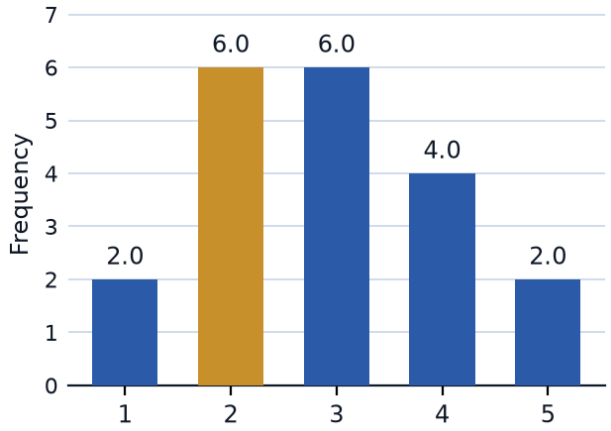
Find the amplitude and period of $y = -4\cos\left(\frac{\pi x}{3}\right)$.

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the distance from $(1,\ -2)$ to $(5,\ 1)$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

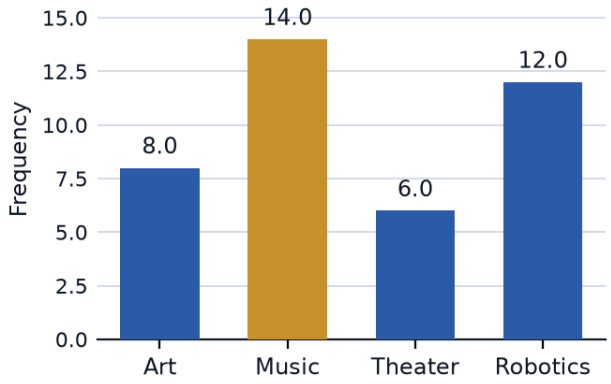
The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars? Show your complete algebraic work.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

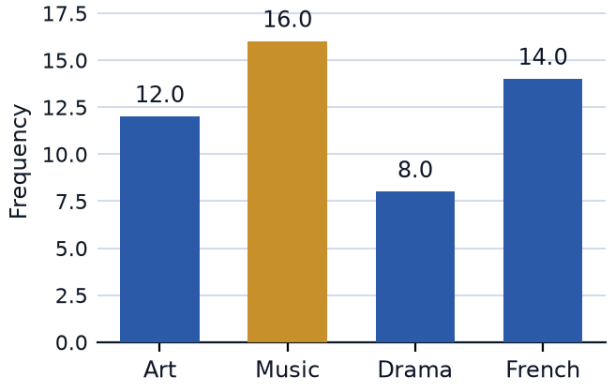
The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music? Show your complete algebraic work.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Advanced Number Theory

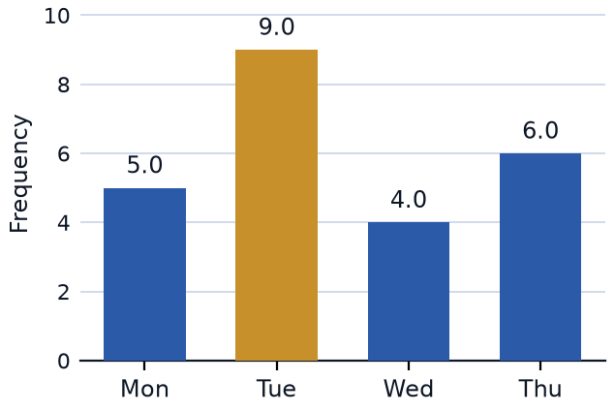
The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama? Show your complete algebraic work.



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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Advanced Algebra & Systems

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days? Show your complete algebraic work.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Advanced Functions

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse? Show your complete algebraic work.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee? Show your complete algebraic work.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Pre-Calculus Concepts

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3:5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally? Show your complete algebraic work.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Of 31 students, 13 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball? Show your complete algebraic work.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A charity raffle sells $1,000$ tickets at 5 each. There is one grand prize of $1,000$ and two second prizes of 500 each. What is the expected net gain for a person purchasing one ticket?

a. -3.00

b. -2.00

c. -1.50

d. 0.00

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random variable X has probability distribution: $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$. What is the expected value $E(X)$?

a. 2.0

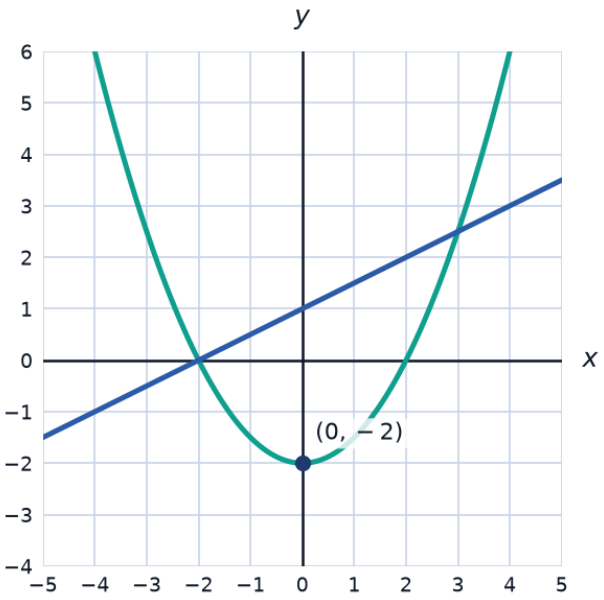
b. 2.1

c. 2.2

d. 2.3

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?



a. 3

b. 5

c. 9

d. 16

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. Approximately what percentage of test takers score between 400 and 600?

a. 50%

b. 68%

c. 95%

d. 99.7%

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student scores 84 on a test with mean 72 and standard deviation 8. What is the student's z-score?

a. 1.25

b. 1.50

c. 1.75

d. 2.00

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a normal distribution with mean 50 and standard deviation 5, what value corresponds to a z-score of -1.6?

a. 42

b. 44

c. 46

d. 48

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

a. 0.50

b. 0.60

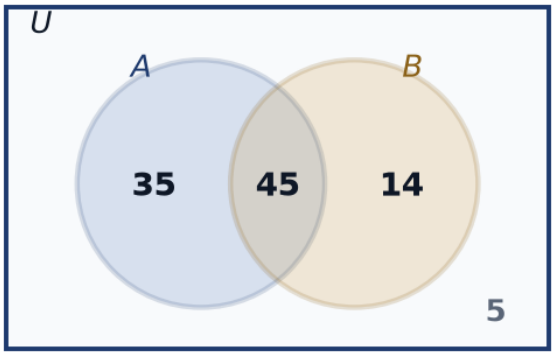
c. 0.75

d. 0.30

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If two events A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?



a. 0.70

b. 0.80

c. 0.90

d. 0.1575

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

a. 0.019

b. 0.020

c. 0.095

d. 0.95

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In how many ways can 6 students be seated in a circle?

a. 120

b. 720

c. 60

d. 24

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

a. 2

b. 3

c. 4

d. 11

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

a. $50,400$

b. $100,800$

c. $302,400$

d. $3,628,800$

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 32 students, 21 play soccer and 10 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{21}{32}$

b. $\frac{10}{11}$

c. $\frac{10}{21}$

d. $\frac{11}{21}$

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 39 students, 19 play soccer and 11 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{8}{19}$

b. $\frac{11}{20}$

c. $\frac{11}{19}$

d. $\frac{11}{39}$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 27 students, 17 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{3}{10}$

b. $\frac{14}{17}$

c. $\frac{3}{17}$

d. $\frac{17}{27}$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 37 students, 29 play soccer and 4 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{25}{29}$

b. $\frac{4}{29}$

c. $\frac{1}{2}$

d. $\frac{4}{37}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Of 36 students, 10 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{5}{18}$

b. $\frac{1}{12}$

c. $\frac{7}{10}$

d. $\frac{3}{10}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

Of 40 students, 10 play soccer and 7 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{7}{30}$

b. $\frac{7}{10}$

c. $\frac{3}{10}$

d. $\frac{1}{4}$

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Of 35 students, 16 play soccer and 9 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{9}{19}$

b. $\frac{9}{16}$

c. $\frac{9}{35}$

d. $\frac{16}{35}$

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Of 34 students, 17 play soccer and 2 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{1}{17}$

b. $\frac{2}{17}$

c. $\frac{15}{17}$

d. $\frac{1}{2}$

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

Of 40 students, 17 play soccer and 5 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{12}{17}$

b. $\frac{5}{23}$

c. $\frac{1}{8}$

d. $\frac{5}{17}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

Of 23 students, 13 play soccer and 5 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{1}{2}$

b. $\frac{13}{23}$

c. $\frac{5}{13}$

d. $\frac{8}{13}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Of 30 students, 13 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{13}{30}$

b. $\frac{1}{10}$

c. $\frac{3}{17}$

d. $\frac{3}{13}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Of 21 students, 12 play soccer and 7 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{5}{12}$

b. $\frac{7}{9}$

c. $\frac{1}{3}$

d. $\frac{7}{12}$

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

Of 40 students, 28 play soccer and 9 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{19}{28}$

b. $\frac{9}{28}$

c. $\frac{3}{4}$

d. $\frac{7}{10}$

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